Middle East Technical University
Department of Electrical and Electronics Engineering

# EE201 – Circuit Theory I

## Lecture 11: Node Voltage Analysis

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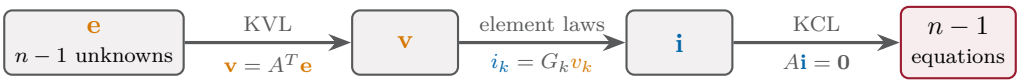
**Colour code used in these notes and in the slides**

current $i$ voltage $v$, node voltage $e$ supernode, nodes Ⓢ

**Motivation.** An LTI resistive circuit can be solved in two ways: by transforming it into a simpler circuit, or by writing its equations as it is. Transforming works for many circuits, but not for all. This lecture gives a systematic way to write the equations of *any* LTI resistive circuit: **node voltage analysis**. Its unknowns are only the $n-1$ node voltages, instead of the $2b$ branch voltages and currents. Mesh current analysis, the same idea with KCL and KVL swapped, comes later.

## 1. The idea: node voltages as the unknowns

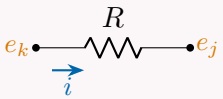
The incidence matrix gives KCL as $A\mathbf{i} = \mathbf{0}$ and KVL as $\mathbf{v} = A^T\mathbf{e}$. So if the node voltages $\mathbf{e}$ are the unknowns, KVL holds automatically. The element laws then give every branch current in terms of $\mathbf{e}$, and only KCL is left to write:




Once $\mathbf{e}$ is found, $\mathbf{v} = A^T\mathbf{e}$ gives the branch voltages and the element laws give the branch currents.

### Node voltage analysis: the procedure

1. Choose a **reference node** (ground); its node voltage is 0.
2. Label the other node voltages $e_1, \ldots, e_{n-1}$.
3. Write **KCL** at every node except the ground, with each current written in terms of node voltages.



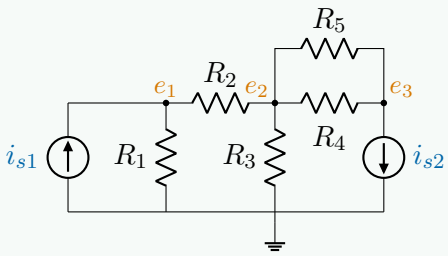

$$i = \frac{e_k - e_j}{R} = G\,(e_k - e_j)$$

The current *leaving* node $k$ through $R$; with the other end at the ground, it is $e_k/R$.

## 2. Resistors and current sources

Start with circuits whose elements are only resistors and current sources. Every branch current is then either known (a source) or a conductance times a difference of node voltages. The circuit of Example 11.1 has $b = 7$ branches, so $2b = 14$ branch voltages and currents. With the bottom node as the ground, only $n - 1 = 3$ node voltages are unknown.

### Example 11.1 — only current sources: KCL at nodes 1, 2 and 3




node 1: $$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

node 2: $$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} + \frac{e_2 - e_3}{R_5} = 0$$

node 3: $$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_2}{R_5} + i_{s2} = 0$$

Every term is a current leaving the node; the source $i_{s1}$ points into node 1, so it enters with a minus sign. With $G_k = 1/R_k$ and the terms collected, the three equations of Example 11.1 become

$$\underbrace{\begin{bmatrix} G_1+G_2 & -G_2 & 0 \\ -G_2 & G_2+G_3+G_4+G_5 & -(G_4+G_5) \\ 0 & -(G_4+G_5) & G_4+G_5 \end{bmatrix}}_{Y_n} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_s}.$$

$Y_n$ is the **node admittance matrix**, and $Y_n\mathbf{e} = \mathbf{i}_s$ are the **node equations**. For a circuit of resistors and current sources, they can be written down directly.

**Writing $Y_n$ by inspection**

- **Diagonal** entry $(k,k)$: the sum of the conductances connected to node $k$.
- **Off-diagonal** entry $(k,j)$: minus the sum of the conductances between nodes $k$ and $j$.
- **Right side**, row $k$: the sum of the source currents entering node $k$.

So $Y_n$ is **symmetric**: the entry $(k,j)$ equals the entry $(j,k)$.

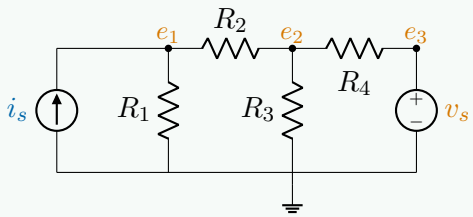
**Why is $Y_n$ symmetric?** Split the columns of $A$ into the resistor branches and the current source branches, $A = [\,A_R \;\; A_S\,]$. KCL reads $A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$. The resistors obey $\mathbf{i}_R = G\,\mathbf{v}_R$ with $G = \mathrm{diag}(G_1,\ldots,G_5)$, and KVL gives $\mathbf{v}_R = A_R^T\mathbf{e}$. Together,

$$\underbrace{A_R\,G\,A_R^T}_{Y_n}\,\mathbf{e} = \underbrace{-A_S\mathbf{i}_S}_{\mathbf{i}_s}, \qquad Y_n^T = A_R\,G^T\,A_R^T = A_R\,G\,A_R^T = Y_n,$$

since $G$ is diagonal. $A_S\mathbf{i}_S$ adds up the source currents leaving each node, so $-A_S\mathbf{i}_S$ adds up those entering it.

## 3. A voltage source to the ground

**Example 11.2 — a voltage source between node 3 and the ground**




The unknowns are $e_1$, $e_2$, $e_3$, so three equations are needed. KCL at nodes 1 and 2:

$$\text{node 1:} \quad -i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

$$\text{node 2:} \quad \frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

**Question.** What does KCL at node 3 give?

**Answer.** The current through $v_s$ is unknown, and it cannot be written with node voltages. So KCL at node 3 gives no useful equation in $e_1$, $e_2$, $e_3$. But the source gives $e_3 = v_s$: this is the third equation.

**Remark: a voltage source to the ground**

If an independent voltage source $v_s$ is connected between the ground and node $k$, then $e_k = v_s$ is known. So $e_k$ is not an unknown, and KCL is not written at node $k$.

**Example 11.2, revisited**

The unknowns are only $e_1$ and $e_2$. Put $e_3 = v_s$ into the two KCL equations:

$$\begin{bmatrix} \frac{1}{R_1}+\frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}.$$

The known $v_s$ moves to the right side, where $v_s/R_4$ acts like a current source into node 2 (a source transformation).

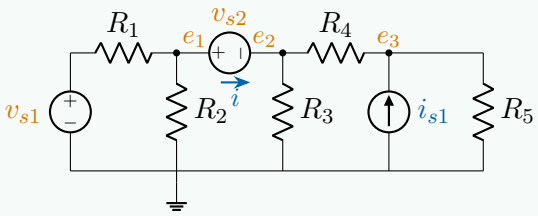
## 4. A voltage source between two nodes: modified node analysis

If a voltage source, independent or dependent, is connected between two nodes and neither is the ground, KCL at these nodes needs its current. This current cannot be written with node voltages: the law of a voltage source fixes its voltage, not its current. **Modified node analysis** offers two ways out.

**Method I**

Let the unknown current $i$ through the voltage source be one more unknown. Write KCL at every node except the ground, and add the law of the source as a **constraint equation**.

**Example 11.3 — Method I**




The top of $v_{s1}$ is at the known voltage $v_{s1}$; $v_{s2}$ is between nodes 1 and 2. Unknowns: $e_1$, $e_2$, $e_3$ and $i$.

node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

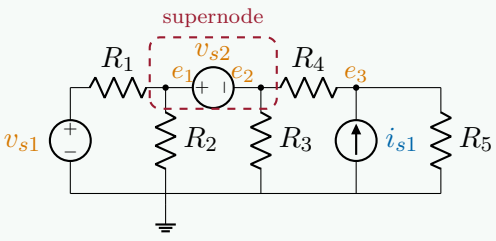
constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

**Method II: the supernode**

Write KCL as if the two nodes of the voltage source were a single node, a **supernode**. Add the law of the source as a constraint equation.

**Example 11.3, revisited — Method II, three unknowns**




supernode: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

constraint: $e_1 - e_2 = v_{s2}$

**Why does it work?** Add the KCL equations of nodes 1 and 2 in Method I: $i$ leaves node 1 and enters node 2, so it cancels. The sum is KCL around the supernode. The current $i$ is no longer needed. In matrix form, with three unknowns and three equations:

$$\begin{bmatrix} G_1+G_2 & G_3+G_4 & -G_4 \\ 0 & -G_4 & G_4+G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}.$$

The matrix is no longer symmetric.

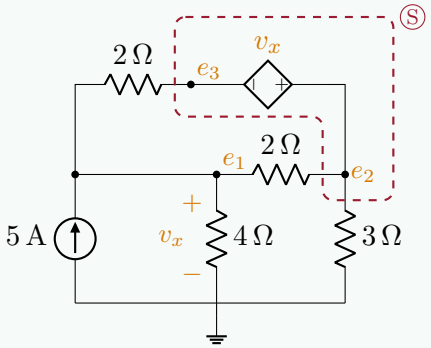
| | **Method I** | **Method II** |
|---|---|---|
| extra unknown | the current $i$ of the source | none |
| KCL written | at every non-ground node | at the supernode and the other nodes |
| extra equation | the constraint (law of the source) | the same constraint |
| size in Example 11.3 | $4 \times 4$ | $3 \times 3$ |
| the current $i$ | part of the solution | from a KCL afterwards |

Both methods give the same node voltages.

## 5. Dependent sources

A dependent source is treated like an independent source of the same kind. Its control variable is written with node voltages; this gives one more constraint equation.

**Example 11.4 — a dependent voltage source**




**Question.** Find the power absorbed by the 3 Ω resistor.
The source $v_x$ is between nodes 3 and 2, and neither is the ground. So use a supernode Ⓢ around nodes 2 and 3. The unknowns are $e_1$, $e_2$, $e_3$.

(1) node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0$$

(2) Ⓢ: $$\frac{e_3 - e_1}{2} + \frac{e_2 - e_1}{2} + \frac{e_2}{3} = 0$$

**Constraint.** The source gives $e_2 - e_3 = v_x$, and its control $v_x$ is the voltage of the 4 Ω, so $v_x = e_1$:

$$e_2 - e_3 = e_1. \qquad (3)$$

**Solving Example 11.4.** Put $e_1 = e_2 - e_3$ into (1) and (2):

$$\tfrac{5}{4}e_1 - \tfrac{1}{2}e_2 - \tfrac{1}{2}e_3 = 5 \;\Rightarrow\; \tfrac{3}{4}e_2 - \tfrac{7}{4}e_3 = 5, \qquad -e_1 + \tfrac{5}{6}e_2 + \tfrac{1}{2}e_3 = 0 \;\Rightarrow\; e_3 = \tfrac{1}{9}e_2.$$

Then $\frac{3}{4}e_2 - \frac{7}{36}e_2 = \frac{20}{36}e_2 = 5$, so $e_2 = 9\,\text{V}$, $e_3 = 1\,\text{V}$ and $e_1 = 8\,\text{V}$. The 3 Ω is between node 2 and the ground:

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\text{W}.$$

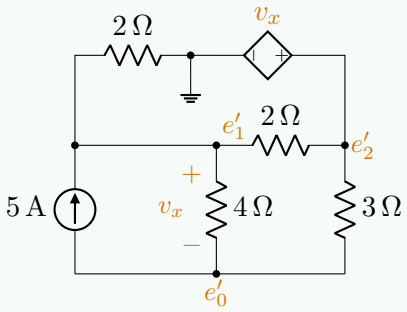
**Remark: a dependent voltage source to the ground**

If a dependent voltage source is connected between node $k$ and the ground, do not write KCL at node $k$: its current is not known. Write the constraint that the dependent source gives instead.

## 6. Choosing a different ground

Any node can be the ground. Changing it shifts every node voltage by the same amount, so the branch voltages, the currents and the powers do not change.

**Example 11.5 — Example 11.4 with node 3 as the ground**




**Question.** Redo Example 11.4 with node 3 as the ground. Does the answer change?
Now $v_x$ is between node 2 and the ground: no supernode is needed, and by the remark above, KCL is not written at node 2. The unknowns are $e_1'$, $e_2'$ and $e_0'$, the voltage of the old ground.

node 1: $$-5+\frac{e_1'}{2}+\frac{e_1'-e_0'}{4}+\frac{e_1'-e_2'}{2}=0$$

node 0: $$5+\frac{e_0'-e_1'}{4}+\frac{e_0'-e_2'}{3}=0$$

constraint: $$e_2'=v_x=e_1'-e_0'$$

**Solving Example 11.5.** Put the constraint into the two KCL equations: $3e_1'+e_0'=20$ and $-7e_1'+11e_0'=-60$. So $e_1'=7\,\text{V}$, $e_0'=-1\,\text{V}$, $e_2'=8\,\text{V}$, and

$$P_{3\Omega}=\frac{(e_2'-e_0')^2}{3}=\frac{9^2}{3}=27\,\text{W},\quad \text{the same.}$$

Every node voltage dropped by $e_3=1\,\text{V}$: $8\to7$, $9\to8$, $0\to-1$.

**Tip.** Put the ground at one end of a voltage source. Then its other node voltage is known, or a supernode is avoided.

## Summary

| In the circuit | What to do |
|---|---|
| resistors and current sources | KCL at every non-ground node: $Y_n\mathbf{e}=\mathbf{i}_s$, written by inspection; $Y_n=A_RGA_R^T$ is symmetric |
| a voltage source from the ground to node $k$ | $e_k=v_s$ is known; no KCL at node $k$ |
| a voltage source between two nodes | Method I: its current as an unknown, plus the constraint; Method II: KCL at the supernode, plus the constraint |
| a dependent source | write its control variable with node voltages: a constraint equation |
| the choice of ground | any node; the branch voltages and currents do not change |

After the node voltages are found: $\mathbf{v}=A^T\mathbf{e}$, then the branch currents from the element laws.